Manifold Ranker

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***Abstract*— Assessing the quality of telecommunications services across municipalities with heterogeneous structural, geographic, and socioeconomic conditions requires methodologies that go beyond simple weighted averages. This paper introduces a Graph Neural Network (GNN)-based manifold ranking framework that learns a latent representation of municipality-level telecom quality indicators and produces a unified ranking score. The proposed architecture employs Residual Graph Convolutional layers operating on a k-nearest-neighbor graph constructed from 19 infrastructure and quality-of-service indicators monitored by the Brazilian National Telecommunications Agency (ANATEL). The model is trained with a multi-objective loss function that simultaneously enforces pairwise ranking consistency, input reconstruction fidelity, geometric smoothness on the learned manifold, and directional alignment with domain knowledge. A monotonicity constraint is imposed on the scoring head to guarantee that improvements in underlying indicators always translate into higher scores. We validate the approach on a longitudinal dataset covering 4,088 Brazilian municipalities over 30 monthly periods (July 2022–December 2024), totaling over 82,000 municipality–period observations. Ablation studies across six experimental configurations demonstrate the contribution of each loss component. Temporal stability analyses via spaghetti plots reveal that the manifold ranker produces smoother and more consistent rankings than the regulatory baseline (IQS), while geographic visualizations confirm alignment with known infrastructure patterns. The framework provides a principled, data-driven alternative to expert-defined composite indices for regulatory quality assessment at national scale.**

***Keywords*— Learn to Rank (LTR), Manifold Learning, Pareto Optimization, Multi Objective Optimization, Quality of Service (QoS)**

I. Introduction

Ranking municipalities in terms of telecommunications Quality of Service (QoS) poses a challenge to regulatory agencies. The ranking is useful not only for diagnostic purposes but for technical and economical evaluation of a given municipality or region.

Defining a ranking strategy is not trivial. One of the problems raised by this challenge are the absence of stablished ground-truth on which cities are better or worse, therefore, evaluating models results depends on contextual knowledge rather than objective comparison. Also, selecting the essential variables to make this analysis is not clear, not only QoS indicators are required to define the strategy but also covariates such as infrastructure and Quality of Experience (QoE) indicators.

Currently, regulatories agencies uses linear composites to stablish QoS scores that are used to ranking municipalities. Brazilian National Telecommunications Agency (ANATEL) defined the Service Quality Index (IQS) through Reference Values Document (DVR), that encapsulates QoS metrics in single valued score to define the telecommunications quality for each municipality.

This pointwise approach lacks a gobal overview of the performance on the municipalities. By not considering a dominance from one over another entities, the ranking became faulty, with scores indistinguishable one from another. Similar entites are attributed near identical values, and the ocurrences are cluttered on narrow regions of the support domain.

To improve distinguishability, one strategy that can be adopted is to compare features from two municipalities and define which one dominates over the other. To find these similarities it is possible to construct a graph, similar entites will be close to each other on the data space.

To asses all the presented challenges, we propose Manifold Ranker, a GNN-based Pareto-inspired model that learns the geometry of the space where the data live in, based on a constructed k-Nearest Neighbors (kNN) graph, and by entity pairwise competition regresses a score value s_i for each municipalities i.

The remainder of the paper is organized as follows. Section II presents the data used in this work and the currently calculated scores. Section III explain the concepts using throughout the development of the methodology applied and presents the developed Manifold Ranker model. Section IV presents the achieved results and a comparison with the currently applied strategies, and discusses them. Section V make some final remarks to the presented work.

II. Dataset and Baseline Creation

For the dataset construction we acquired QoS and QoE indicators, and infrastructure covariates provided by ANATEL, creating a matrix $\mathbf{X}^{\text{raw}} \in \mathbb{R}^{i \times d}$, where i correspond to the index key and d correspond to the number of features. $\mathbf{X}^{\text{raw}}$ is indexed by two keys: *month*, which identifies the observation period, and *municipality*, the official Brazilian Institute of Geography and Statistics (IBGE) numeric code that uniquely identifies each municipality. Each row therefore corresponds to one municipality–month pair.

As QoS features we selected eight features normalized to $[0, 1]$ with higher values denoting superior performance, IND_1 (voice-call connection rate), IND_2 (voice-call drop rate, complemented), IND_3 (data-connection rate), IND_4 (download

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and upload speed compliance rate), IND_5 (data-connection latency, complemented), IND_6 (jitter, complemented), IND_7 (packet-loss rate, complemented), and IND_8 (service availability rate). For QoE feature, it was selected the ISG is the General Satisfaction Index, which is a composite of QoE indicators for a given tuple municipality–period. Infrastructure is represented on the dataset by IBC, which is the Brazilian Coverage Index, and the $\mathrm{INF4}_{\mathrm{UP},\{3\mathrm{G},4\mathrm{G},5\mathrm{G}\}}$ and $\mathrm{INF4}_{\mathrm{DL},\{3\mathrm{G},4\mathrm{G},5\mathrm{G}\}}$ indicators, which records the median download speeds over each mobile data technology: 3G, 4G, and 5G.

To establish a baseline comparison with the currently applied metodologies in regulatory agencies, we present the consolidated calculated scores the legacy IQS and the recently revised IQS. Both of them linearly combines the same QoS features selected to use in the model.

A. Legacy IQS

The Anatel legacy composite, $\mathrm{IQS}_{\mathrm{leg}}$, aggregates the K normalized technical quality-of-service indicators through a fixed regulator-defined convex combination,

$$\mathrm{IQS}_{\mathrm{leg}} = y_{\max} \sum_{k=1}^{K} w_k^{\mathrm{leg}} \, \mathrm{IND}_k, \tag{1}$$

where $y_{\max} = 10$ is the upper bound of the user satisfaction scale, each $\mathrm{IND}_k \in [0,1]$ is a normalized technical indicator with higher values denoting superior performance, and the weight vector $w^{\mathrm{leg}} \in \Delta^{K-1}$ lies on the probability simplex $\Delta^{K-1} = \{w \in \mathbb{R}^K : w_k \geq 0, \sum_k w_k = 1\}$. Under the legacy specification, the weights are fixed at $w^{\mathrm{leg}} = (0.1, 0.1, 0.1, 0.2, 0.1, 0.1, 0.1, 0.2)$, assigning greater relevance to IND_4 and IND_8, while distributing equal weight among the remaining indicators.

B. Revised IQS

The Anatel revised composite, $\mathrm{IQS}_{\mathrm{rev}}$, introduced in 2025 through the updated DVR [**?**], replaces the fixed weights with a performance-dependent scheme combined with a coverage bonus. The indicators are first ranked from worst-performing to best-performing within each municipality, and the weight assigned to rank position r is

$$\alpha_r = \frac{1}{r\, H_K}, \tag{2}$$

where $H_K = \sum_{j=1}^{K} 1/j$ is the K-th harmonic number. The revised composite is then defined as

$$\mathrm{IQS}_{\mathrm{rev}} = y_{\max} \sum_{r=1}^{K} \alpha_r \cdot \mathrm{IND}_{\pi(r)} + B_{\mathrm{cov}}, \tag{3}$$

where π is the permutation ordering indicators from worst to best performance, and B_{cov} is a coverage bonus corresponding to a fixed increment of five points, added whenever coverage with 4G or higher technology in the municipality's urban area reaches at least 95%. When $B_{\mathrm{cov}} = 0$, the revised rule reduces to a linear function of the indicators; when $B_{\mathrm{cov}} \neq 0$, it becomes affine. The inverse-harmonic weighting assigns greater importance to the worst-performing indicators, penalizing municipalities with more critical technical dimensions, in contrast to the static weight distribution of the legacy composite.

III. Methodology

Fig. 1 presents an overview of the pipeline process, which will be detailed as follows:

A. Preprocessing

By exploring the selected data that consists the dataset, we verify that it is composed of heavy-tailed, heterogeneous data. Aiming to transform the data with that characteristics into a suitable representation for neural optimization we define a preprocessing step.

To reduce the dynamic range of the features that follows power-law or log-normal-like distributions, common among infrastructure data we apply a Log1p transformation, shown in Eq. 4, for numerical stability this transformations prevents the $\log(0)$ by adding one to the paramater.

$$\mathbf{X}^{\log} = \log(1 + \mathbf{X}^{\mathrm{raw}}). \tag{4}$$

To reduce outliers we apply a Robust scaler, which filter only the date within an interquantile range selected to be $\{0.05, 0.095\}$, Eq. 5 shows the processing.

$$\tilde{x}_{i,j} = \frac{\mathbf{x}_{i,j}^{\log} - \mathrm{median}(\mathbf{x}_{i,j}^{\log})}{Q_{0.95}(\mathbf{x}_{i,j}^{\log}) - Q_{0.05}(\mathbf{x}_{i,j}^{\log})}. \tag{5}$$

From the robust-scaled features, we apply a Standar scaling, Eq. 6 centering the variables into zero mean and unit variance, conditioning for first-orders optimizers, and making

$$\mathbf{X}_{i,j}^{\mathrm{model}} = \frac{\tilde{x}_{i,j} - \mu_j}{\sigma_j}, \tag{6}$$

where μ_j and σ_j are the mean and the standard deviation for the j feature across all municipalities.

Additionally from the $\mathbf{X}^{\mathrm{model}}$, we create a proxy matrix $\mathbf{X}^{\mathrm{pareto}}$ that serves as support for the model training serving as a "compass" general directional guide. $\mathbf{X}^{\mathrm{pareto}}$ is constructed as shown in Eq. 7, by Min-Max scaling the $\mathbf{X}^{\log}$

$$\mathbf{X}_{i,j}^{\mathrm{pareto}} = \frac{\mathbf{x}_{i,j}^{\log} - \min_i \mathbf{x}_{i,j}^{\log}}{\max_i \mathbf{x}_{i,j}^{\log} - \min_i \mathbf{x}_{i,j}^{\log}} \in [0,1], \tag{7}$$

Then, a scalar proxy utility is defined as the mean across features

$$u_i = \frac{1}{D} \sum_j \mathbf{X}_{i,j}^{\mathrm{pareto}}, \tag{8}$$

this proxy is not a true Pareto ranking; it is a heuristic that proivdes a consistent direction for training in the abscence of labeled rankings.

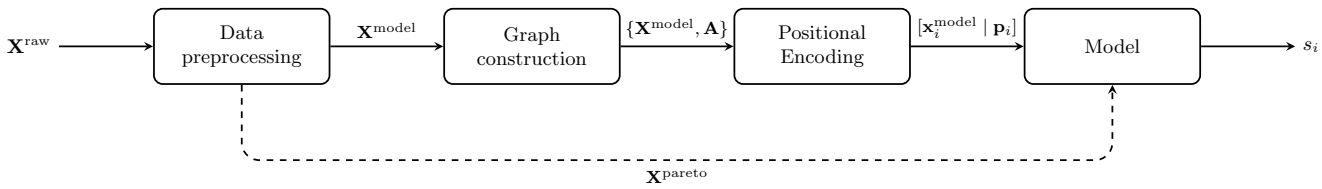


Fig. 1: Pipeline for the proposed methodology

B. Graph Construction and Manifold Structure

Graphs are mathematical structures capable extracting relationship from data. These relationship enables partial property sharing, having local properties. Geometrically we can visualize this as an estimation of a manifold.

We selected the kNN as the method for constructing the graph. This method calculates the distance between each node and select the k nearest neighbors by using a distance metric, Minkowski in this case, as a result we get a index matrix with the nearest neighbors for each node $\mathcal{I} \in \{1, ..., N\}^{N \times k}$, with N being the number of nodes.

Most implementations of kNN algorithm traces the most nearest neighbor as the node itself, distance zero from itself, i.e., $i \to i$, creating a loop in the graph. This loop brings a residual glance to the model, that can be explored to prevent information vanishing.

C. Positional Encoding

Message-passing Graph Neural Networks (GNN)s inherently possess a permutation-invariance property, which implies that relabeling nodes does not change the resulting computations. Due to this characteristic, the network lacks a notion of global coordinates on the dataset manifold. To inject this structural awareness into the architecture, we utilize specific methods to extract positional encodings $p_i \in \mathbb{R}^{d_{pe}}$.

To obtain these encodings we fall back to Principal Component Analysis (PCA), which finds the orthogonal directions of maximum variance and projects the data onto these principal components, generating a linear approximation of the manifold's global coordinates. Alternatively, Spectral Embedding could be used as a more robust mechanism for extracting non-linear coordinates when the graph dimension respects processing constraints.

The computed features p_i are subsequently processed using a Standard Scaler. This normalization is crucial because p_i is directly concatenated with the node's original data x_i, thereby averting any scale dominance that could occur if the positional coordinates carried disproportionately larger magnitudes.

D. Model Architecture

The core processing logic is built upon message-passing principles, where at each layer ℓ, a node representation $h_i^{(\ell)}$ is successfully updated through aggregating features coming from interacting neighbors:

$$m_i^{(\ell)} = \text{AGG}\left(\left\{\phi\left(h_j^{(\ell)}\right) : (j \to i) \in E\right\}\right), \quad h_i^{(\ell+1)} = \psi\left(h_i^{(\ell)}, m_i^{(\ell)}\right), \tag{9}$$

where ϕ provides a linear learnable transformation, AGG is a permutation-invariant aggregation function (mean pooling in this case), and ψ resolves the aggregated messages with the previous node state.

Following these concepts, our network leverages a Residual Graph Convolutional Network (ResGCN) block for ψ, incorporating skip-connections across sequential computations. This layout not only offers direct paths reducing gradient vanishing but also systematically mitigates typical oversmoothing found in deep node propagations, thus preserving essential local node characteristics. Due to the rapid computational expansion related to neighborhood scaling, bounded graph subsampling is also used to filter out distant connections, preventing unmanageable memory footprints during training and inference phases.

All node activations apply the Gaussian Error Linear Unit (GELU) operators, achieving a higher level of continuous differentiable non-linearity to map node variations effectively. Furthermore, these non-linear elements coupled with the model components inherently satisfy monotonicity constraints, confirming that positive variations across individual QoS metric values consistently translate to non-decreasing final evaluation rankings.

At its endstage, the model functions as a multi-task component divided into two independent heads mapping from the final representations. A reconstruction branch serves as a local decoder measuring deviations mapped against the original scaled inputs, while the main ranking head resolves latent combinations over a sequence of Multi-Layer Perceptrons enforcing the ultimate scalar mapped output bounded in the interval $s_i \in [0, 1]$.

E. Training Objectives and Loss Terms

In order to optimize accurately across several perspectives, the entire system adopts a weighted sum composing four unique loss elements focused on distinct architectural attributes, expressed as:

$$\mathcal{L} = \lambda_{\text{rank}}\mathcal{L}_{\text{rank}} + \lambda_{\text{recon}}\mathcal{L}_{\text{recon}} + \lambda_{\text{geo}}\mathcal{L}_{\text{geo}} + \lambda_{\text{compass}}\mathcal{L}_{\text{compass}}. \tag{10}$$

The Compass Loss focuses on guiding the global representation direction strictly oriented towards a defined proxy vector u_i. It applies the Mean Squared Error directly targeting the outputs computed across predictions s_i available within a batch B, driving parameter consistencies matching optimal general directional improvements:

$$\mathcal{L}_{\text{compass}} = \frac{1}{|B|}\sum_{i \in B}(s_i - u_i)^2. \tag{11}$$

The Reconstruction Loss is computed similarly tracking properties emerging from the alternate decoder head. It evaluates the reconstructed parameters $\hat{\mathbf{x}}_i$ against the model input features $\mathbf{x}_i^{\text{model}}$, establishing a strong autoencoder normalization constraint counteracting degenerative multi-dimensional reductions:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|B|} \sum_{i \in B} \|\hat{\mathbf{x}}_i - \mathbf{x}_i^{\text{model}}\|_2^2. \tag{12}$$

The Pairwise Ranking Loss establishes structural hierarchies via pairwise ordering comparisons inside drawn batch combinations using margin parameter constraints m. Assuming target relation labels computed as $y = \text{sign}(u_A - u_B)$, it effectively captures and orients relative quality improvements directly from pair combinations (A, B):

$$\mathcal{L}_{\text{rank}} = \frac{1}{|B|} \sum_{(A,B) \in B} \max(0, m - y(s_A - s_B)). \tag{13}$$

Finally, the Geometric Smoothness Loss focuses on restricting utility magnitude variations over sub-edge neighbor spaces E_{sub} mapping structurally linked municipalities. Derived from the discrete graph Dirichlet formulation, this term asserts that similar components share adjacent values preventing abrupt rank variations representing spatial smoothness:

$$\mathcal{L}_{\text{geo}} = \frac{1}{|E_{\text{sub}}|} \sum_{(i,j) \in E_{\text{sub}}} (s_i - s_j)^2. \tag{14}$$

IV. Results and Discussion

In this section, we present the evaluation of the proposed Manifold Ranker framework against the regulatory baseline composites.

Fig. 2 illustrates the probability distributions for the baseline indicators IQS_{leg}, IQS_{rev}, and the inferred utility score s. The histogram clearly demonstrates that the IQS composites are heavily clustered near the upper bound of the scale (around 0.9), with the legacy composite (IQS_{leg}) being closer to 1.0 and the revised indicator (IQS_{rev}) demonstrating a slightly smaller peak. This significant score aggregation near the maximum value reflects a loss of global distinction when evaluating similarly performing municipalities. In contrast, the utility score computed by the proposed GNN model is widely spread over the domain, presenting a bimodal behavior. The distribution's largest mode is located near 0.6, with a secondary mode clustering near 0.8. This structural spacing explicitly favors rank discrimination, indicating that the multi-objective optimization successfully captures the diverse technical variations between the observations.

To spatially validate this framework, we consider a single evaluation frame representing the period of December 2024. Fig. 3 shows the comparative geographic mapping for each metric across the Brazilian municipalities. The map visualization strictly corroborates the patterns noticed in the distribution histogram. While the GNN utility score exhibits well-distributed continuous performance transitions smoothly highlighting distinct geographic regions, the traditional indicators cluster the vast majority of points into the highest grading tiers. Such behavior emphasizes the network's capability of mapping dynamic multi-dimensional representations without artificially saturating the score margins.

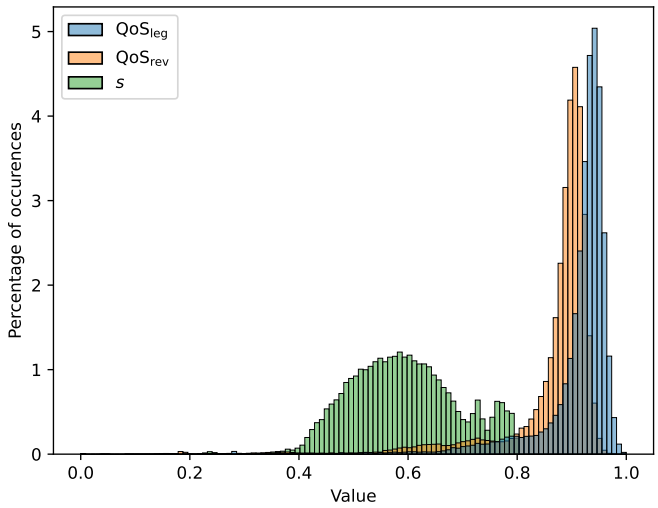
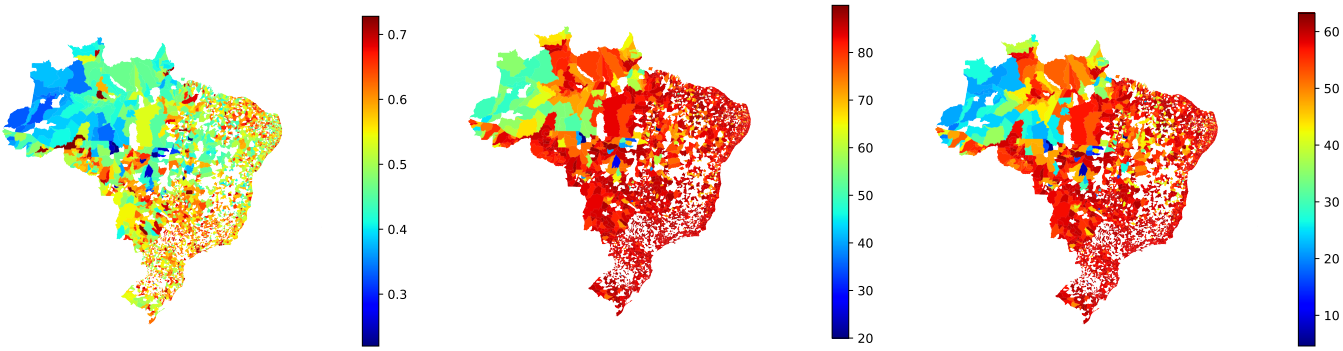


Fig. 2: Histogram of IQS_{leg}, IQS_{rev} and s scores.

V. Conclusions

In this paper, we introduced Manifold Ranker, a GNN-based framework designed to resolve ranking imbalances in telecommunications quality assessment. While current regulatory composite metrics, such as the legacy and revised IQS, suffer from severe density clustering limiting spatial and technical discrimination, our proposed model effectively maps structural manifold relationships. By combining message-passing aggregations with a multi-objective loss function consisting of pairwise ranking, feature reconstruction, geometric smoothness, and a directional compass penalty, the architecture evaluates a consistent and continuous scoring metric. As demonstrated by the bimodal score distributions and spatial map visualizations, the framework yields well-distributed and distinguishable municipality rankings, overcoming traditional saturation issues.

For future work, we plan to robustly benchmark the proposed approach by comparing its performance against established Learning to Rank (LTR) baselines in the literature. Furthermore, there is significant potential in expanding this analysis to evaluate diverse settings across other socioeconomic domains. Finally, exploring alternative positional encoding methods to evaluate different topological representations, alongside testing other activation functions to assess deep structural stabilities, remain as planned efforts for subsequent research steps.



(a) Manifold Ranker s (b) $\mathrm{IQS}_{\mathrm{leg}}$ (c) $\mathrm{IQS}_{\mathrm{rev}}$

Fig. 3: Brazilian map visualization of scores for $\mathrm{IQS}_{\mathrm{leg}}$, $\mathrm{IQS}_{\mathrm{rev}}$ and s.